



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
JANUARY 2024**

Course : EC3114 (Software Modeling and Analysis)

Time: 8.00AM – 11.00AM
(3 hours)

Lecturer : Ts. Harlina Harun

Date: 24 May 2024

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **2** printed pages.
(excluding the front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

- 1 a. Draw a state diagram that describes the lifetime of a DVD in a DVD rental store. You can assume that a DVD is purchased, packaged properly (plastic case with identification information on the outside), put in the DVD store database, and is then put up for rental. Customers, who choose to rent it, check it out and return it in 3 days. If a customer fails to return it, the store calls him/her the next day. The call is repeated a second time after 2 more days, and if the DVD is not returned within 2 more days, the store delegates the matter to a collection agency and removes the DVD from its collection. If the DVD is damaged on return, it is removed from the collection database also. Finally, if the DVD is missing during the annual store inventory, it is removed from the collection database as well. **(15 MARKS)**

Make sure to define events, conditions, and actions for transitions in your diagram, where appropriate

- b. To buy a book electronically from chapters.com, a customer needs to select the book from a list provided by Chapters' ecommerce system, provide credit card information to the system, and then the system gets authorization from the bank for the payment, and if positive confirms the sale. The order is then sent to the orders department and when the book becomes available, it is shipped to the customer. Also, the order department charges the customer's credit card by informing the bank of the amount. **(10 MARKS)**

Draw a sequence diagram that models this process. Make sure to model all relevant actors and the interactions between them. Do show explicitly the time intervals when different actors actively participate in the process you are modeling.

- 2 a. Listed below are some investigation techniques that can be used when analyzing a system. For each, give a scenario situation, or circumstance where that technique may not be suitable or effective even if information is available and you have access to users. **(10 MARKS)**
- I. Document review
 - II. Interview
 - III. Observation
 - IV. Document sampling
 - V. Questionnaire

- b. Explain the difference between structural modelling, functional modelling and behavioural modelling. **(3 MARKS)**
- c. IT projects may fail for one or more reasons. Give **THREE (3)** reasons for failure in product quality and **THREE (3)** reasons for failure in IT team productivity. **(12 MARKS)**
- 3 a. Discuss **SIX (6)** different types of stakeholders who might be interested in new system development and could help in providing requirements. **(12 MARKS)**
- b. Differentiate between functional and non-functional requirements. Write **TWO (2)** examples of each requirement. **(8 MARKS)**
- c. List any **FIVE (5)** Object Oriented Modelling. **(5 MARKS)**
- 4 a. Discuss any **SIX (6)** roles of members of the software development team who are involved in a new system **(12 MARKS)**
- b. Discuss **THREE (3)** types of scenarios in use case analysis **(9 MARKS)**
- c. Explain **FOUR (4)** objectives of requirement analysis. **(4 MARKS)**

-END OF QUESTION PAPER-